Ex.No.5

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WRITE A PYTHON PROGRAM TO PERFORM NETWORK SCANNING USING SCAPY

AIM:

To perform Network Scanning using Scapy module in python.

ALGORITHM:

STEP 1: Start the program & Download and install NpCap.
STEP 2: Import the required modules (ARP, Ether, srp) from the Scapy library.
STEP 3: Read the target network address (e.g., 192.168.1.0/24) from the user.
STEP 4: Create an ARP request packet with the entered network as the destination.
STEP 5: Create an Ethernet frame with the broadcast MAC address (FF:FF:FF:FF:FF:FF).
STEP 6: Combine the Ethernet frame and ARP request to form a complete packet.
STEP 7: Send the packet to the network using the srp() function and receive the responses.
STEP 8: Display the table headers: IP Address and MAC Address.
STEP 9: Iterate through each received response: Extract the sender's IP address (psrc) and sender's MAC address (hwsrc).
STEP 10: Display both addresses.
STEP 11: Stop the program**.**

CODING

```
from scapy.all import ARP, Ether, srp
network = input("Enter network address : ")

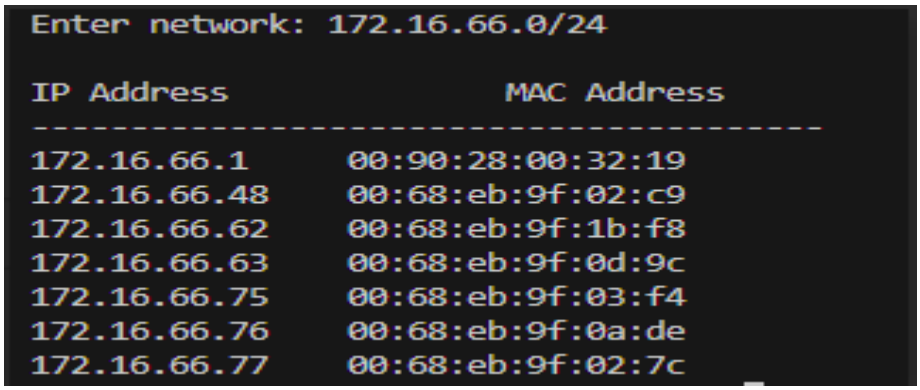
arp = ARP(pdst=network)
ether = Ether(dst="ff:ff:ff:ff:ff:ff")
packet = ether / arp

result = srp(packet, timeout=2, verbose=False)[0]

print("\nIP Address\t\tMAC Address")
print("-" * 40)

for _, received in result:
    print(f"{received.psrc}\t{received.hwsrc}")
```

OUTPUT



RESULT:

Thus, the program to perform Network Scanning using Scapy module in python is executed successfully.